USB Blaster installation tutorial (using **Quartus II 13.0** as an example)

（1） When connecting to the PC for the first time, "Installing device driver software" will appear on the PC side, as shown in Figure 1. Of course the search for the driver software will not be successful, after all the USB Blaster downloader driver is not provided by Windows. The user can wait for a while until the search fails or force the setting "Skip getting driver software from Windows Update" and go to manual setup.



Figure 1 USB Blaster driver search dialog

（2） At this point, open My Computer → Management → Device Manager and you will see the USB-Blaster device marked with a yellow question mark in Other Devices, as shown in Figure 2:



Figure 2 USB Blaster device without drivers installed

（3） Select the USB-Blaster device, right click, select "Update Driver Software" and select "Browse Computer for Driver Software", as shown in Figure 3

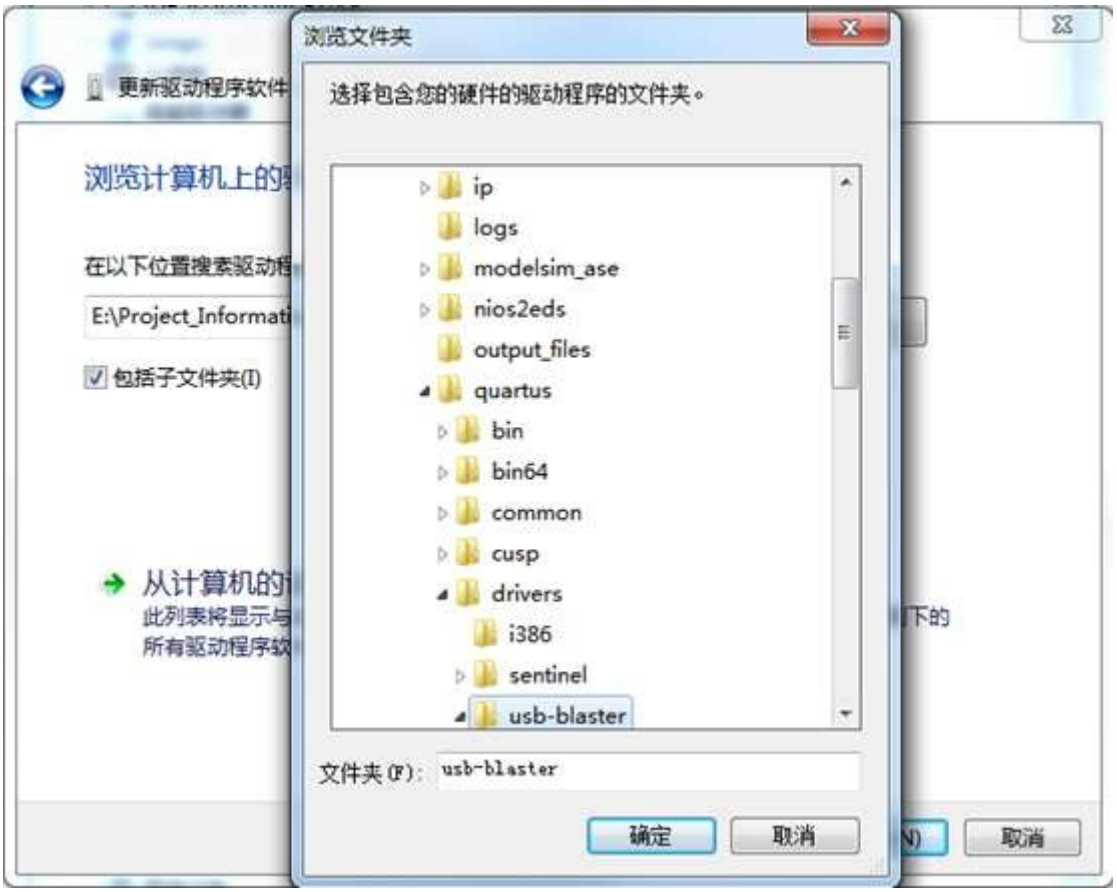


Fig. 3 Select the Manual Find Driver option

（4） Browse to the Quartus II software installation directory and select D:\altera\13.0\quartus\drivers\usb-blaster (depending on the user's software installation directory, this may not be the same here), as shown in Figure 4 and Figure 5.

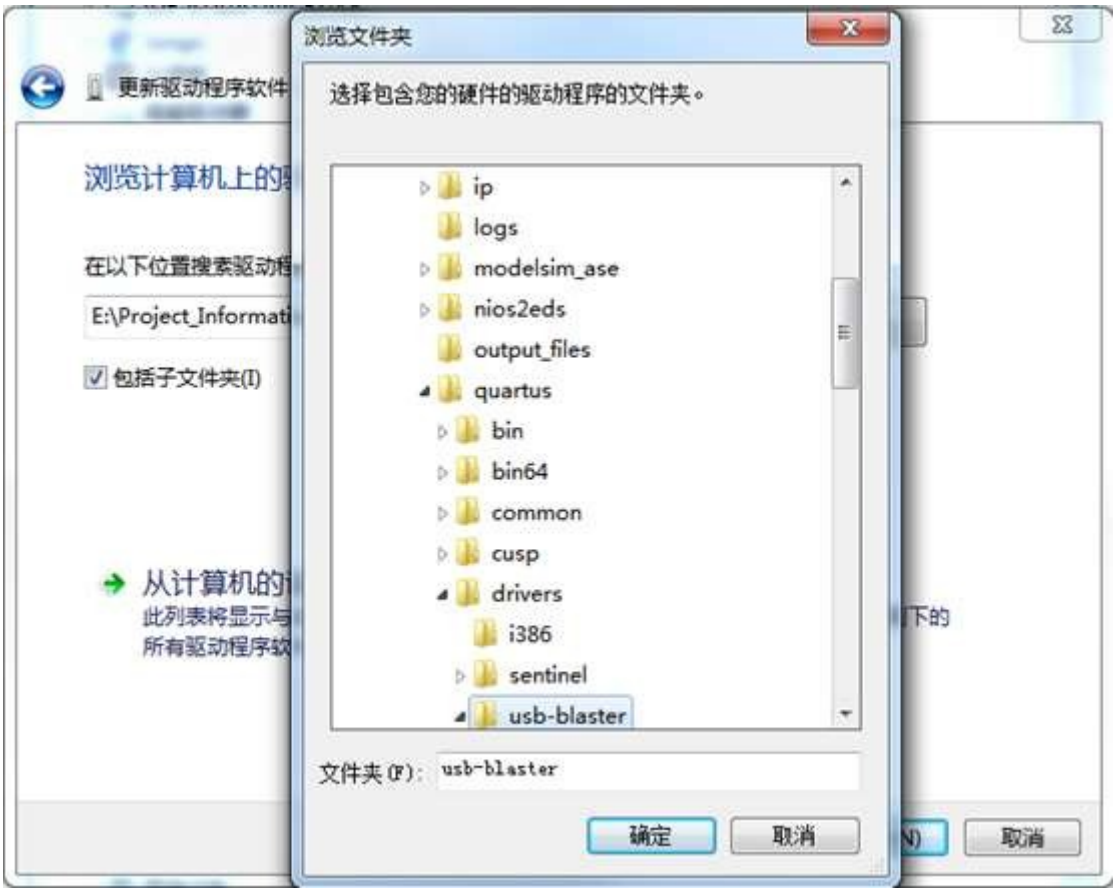


Figure 4 USB Blaster driver directory designation 1

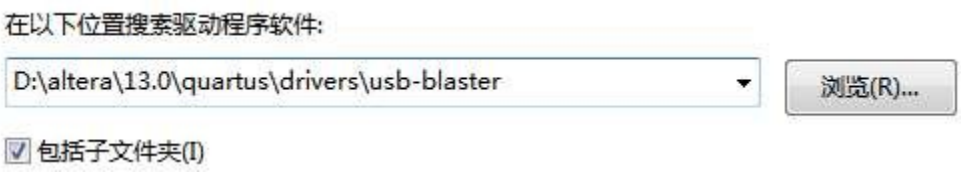


Figure 5 USB Blaster driver directory designation 2

（5）Click OK and wait for the USB Blaster driver to finish installing, as shown in Figure 6.

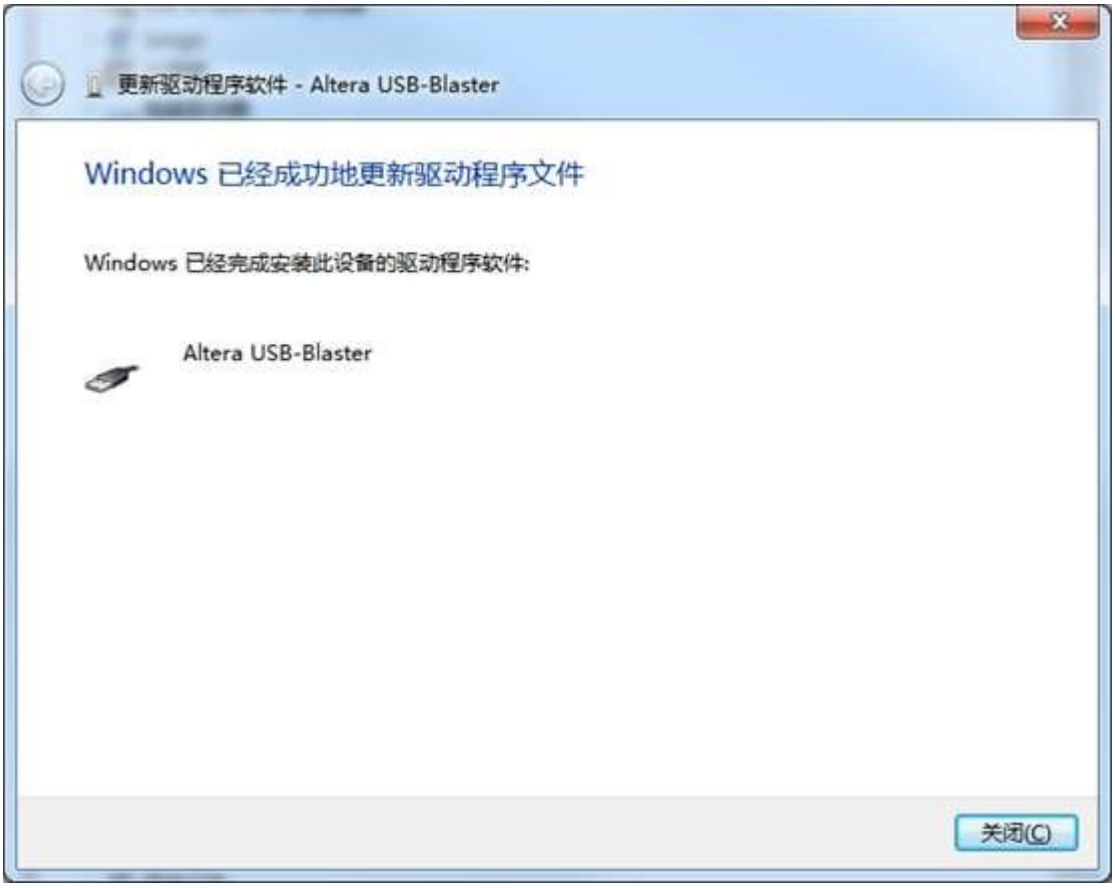


Figure 6 USB Blaster driver installation complete

（6）At this point, back in the Device Manager, the user can locate the Altera USB-Blaster device in the Universal Serial Bus Controller, as shown in Figure 7.



Figure 7 Altera USB Blaster driver recognition interface

At this point, we have completed the installation of the USB Blaster downloader driver and can now configure and debug the FPGA from the computer.

Installation Possible 🌩 Presenting Problems

I. The installation of the driver shows that a digital signature is required.

Due to the problem of Win 10 system authentication, individual software requires digital signature when installing Win 10 system. However, QuartusII software does not have digital signature, so we need to manually turn off the forced digital signature function of Win 10 system.

2. the device manager has installed the drive and recognized it normally, but there is no USB Blaster option in the download page of the Quartus software, there may be several reasons:

1) Causes of the problem: The USB Blaster driver version and the Quartus software version do not correspond to each other.

The solution: The Quartus software will automatically generate the driver when it is installed, just find the driver in the Quartus installation directory and install it, do not use other versions of the USB Blaster driver. The USB-Blaster driver is in folder "\Altera\version number\Quartus\drivers\usb-blaster".

2) System problem.

The solution: Open Device Manager, click Services, find Altera JTAG Server, and right click to restart.

3) System problem.

The solution: Select Altera USB Blaster in Device Manager and tap right to disable it, then tap right to start it again.

4) Digital signature problem, solved according to method 1.